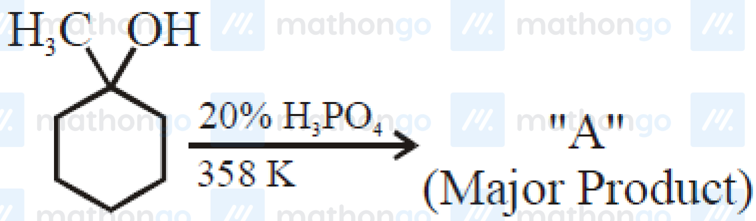
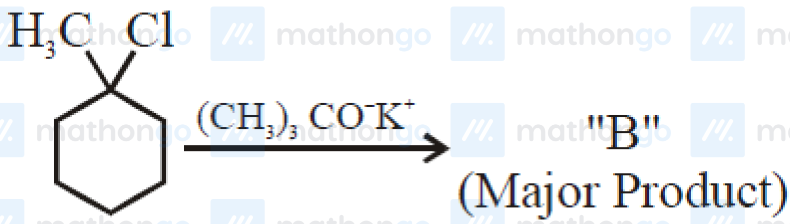
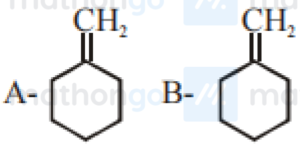
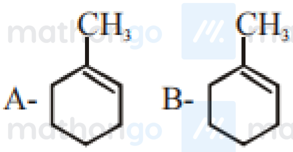
Q1: 16 March (Shift 1) - Single Correct

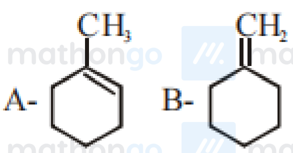


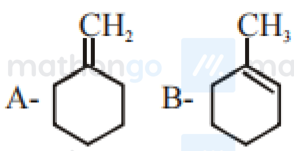


The product "A" and "B" formed in above reactions are

(1) 

(2) 

(3) 

(4) 

Q2: 16 March (Shift 2) - Single Correct

Ammonolysis of Alkyl halides followed by the treatment with NaOH solution can be used to prepare primary, secondary and tertiary amines. The purpose of NaOH in the reaction is:

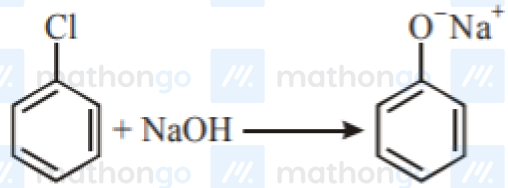
(1) to remove basic impurities

(2) to activate NH_3 used in the reaction

(3) to remove acidic impurities

(4) to increase the reactivity of alkyl halide

Q3: 17 March (Shift 1) - Single Correct



The above reaction requires which of the following reaction conditions?

(1) 573 K, Cu, 300 atm

(2) 623 K, Cu, 300 atm

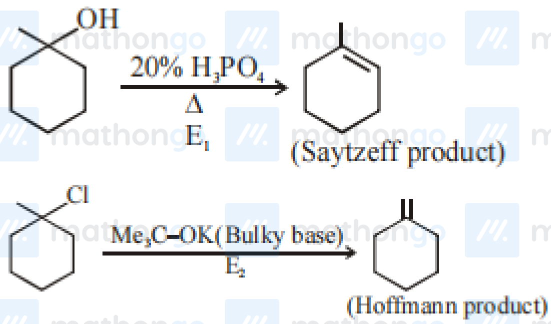
(3) 573 K, 300 atm

(4) 623 K, 300 atm

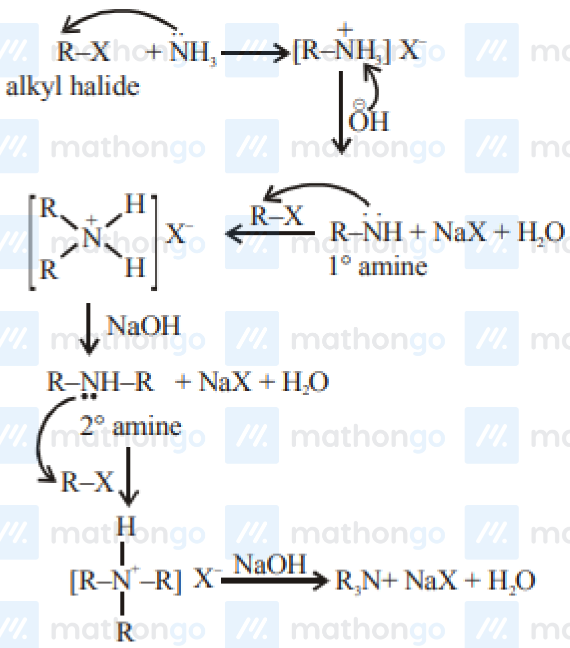
Answer Key

Q1 (3) **Q2** (3) **Q3** (4)

Q1 (3)

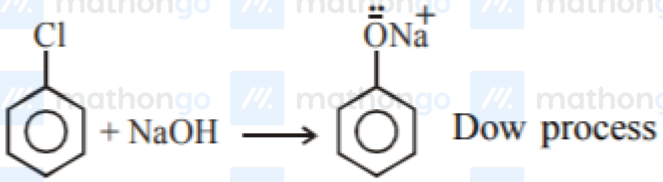


Q2 (3)



So the purpose of NaOH in the above reactions in to remove acidic impurities.

Q3 (4)



Temperature $= 623$ K

Pressure $= 300$ atm